

Refined 3D Index — Worked Example

m022

Generated by `manifold-index`

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1 Setup

We walk through the full refined-index computation for the census manifold `m022`. Every intermediate quantity below is the value actually produced by `manifold.index.core`; the layout is pedagogical so a reader can reproduce any step by hand.

Manifold	m022
Volume	2.98912...
Tetrahedra n	4
Cusps r	1
Internal edges $n - r$	3 (3 easy, 0 hard)
Series order	$q^{20/2}$ ($N_{\max} = 10$)
Display sectors (per cusp)	$(\alpha, \beta) \in \{(0, 0), (0, -\frac{1}{2}), (\frac{1}{2}, 0), (0, -1), (1, 0)\}$

Notation. Every displayed index carries the manifold name as a subscript, as in the in-app series table. An unfilled sector reads $\mathcal{I}_{m022}(e\alpha - \frac{m}{2}\beta)$ with α the meridian, β the longitude, and (m, e) the internal charge variables related by $(\alpha, \beta) = (e, -m/2)$. Hard edges (if present) carry fugacities η^{nW_a} (0-indexed in a); residual cusp η 's carry η^{nV_i} . The tetrahedron factor is $\mathcal{I}_{\Delta}(m_a, e_a)$. A Dehn-filled index along user slope $P\alpha + Q\beta$ is written $\mathcal{I}_{(m022)_{P\alpha+Q\beta}}$ — the manifold name moves inside a parenthesised subscript to make room for the slope, matching `build_fill_row_cells`.

A note on m022. The edge classification (Step 2) returns 0 hard edges: every internal edge equation can already be solved for some η . That means m022 carries *no* η -fugacities, and the “refined” index coincides with the ordinary 3-D index. The Weyl condition (Step 6) reduces to the $m \mapsto -m$ symmetry of the pure q -series. For a manifold with non-empty hard-edge set, the sectors would additionally carry $\eta^{\pm n W_a}$ structure.

2 Step 1 — Triangulation data

SnapPy reports the gluing equations as a $(n + 2r) \times 3n = 6 \times 12$ integer matrix. Each row encodes exponents $(f_i, g_i, h_i)_{i=1}^n$ of (z_i, z'_i, z''_i) in a relation $\prod_i z_i^{f_i} z_i'^{g_i} z_i''^{h_i} = 1$.

Edge rows ($n = 4$). Columns group in triples (f_i, g_i, h_i) for $i = 1, \dots, 4$.

$$G_{\text{edge}} = \left(\begin{array}{ccc|ccc|ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 & 0 & 2 & 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 0 \end{array} \right)$$

Cusp rows ($2r = 2$): **meridian** α , **longitude** β .

$$G_{\text{cusp}} = \left(\begin{array}{ccc|ccc|ccc|ccc} 2 & -1 & 0 & 1 & 0 & -1 & 0 & -1 & -1 & 0 & -1 & 0 \\ -1 & 1 & 1 & -1 & -1 & 1 & 0 & 2 & 0 & 0 & 2 & 0 \end{array} \right)$$

How to read: the four edge rows state that each ideal edge glues up to total holonomy 1; the two cusp rows give the products that express meridian and longitude around the single cusp.

3 Step 2 — Edge classification

An edge row is *easy* if, after column-wise inspection, at most one of (z_i, z'_i, z''_i) is non-zero per tetrahedron; otherwise *hard*. For m022:

Label	Independent?	Row (original index)
E ₀	✓	edge 0
E ₁	✓	edge 3
E ₂	✓	edge 2

No hard edges are needed, so the hard-padding list is empty; the $n - r = 3$ internal coordinates $e_{\text{int},0}, e_{\text{int},1}, e_{\text{int},2}$ correspond directly to the three independent easy edges $\{E_0, E_1, E_2\}$.

4 Step 3 — Neumann–Zagier data

Symplectic reduction produces $g_{\text{NZ}} \in \text{Sp}(2n, \mathbb{Q})$ of shape 8×8 . Columns are ordered $(Z_1, Z_2, Z_3, Z_4, Z_1'', Z_2'', Z_3'', Z_4'')$, rows are grouped $[\text{meridians}_r \mid \text{hard}_{d_h} \mid \text{easy}_{d_e} \mid \text{longitudes}/2_r \mid \text{int. momenta}_{n-r}]$, here $[1 \mid 0 \mid 3 \mid 1 \mid 3]$.

$$g_{\text{NZ}} = \left(\begin{array}{cccc|cccc} 3 & 1 & 1 & 1 & 1 & -1 & 0 & 1 \\ 2 & 1 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 0 & 1 & -2 & 0 & 1 \\ \hline -1 & 0 & -1 & -1 & 0 & 1 & -1 & -1 \\ -2 & -1 & -1 & -2 & -1 & 1 & 0 & -1 \\ -3 & -1 & -2 & -1 & -1 & 1 & 0 & -1 \\ -2 & 0 & -1 & -1 & -1 & 1 & 0 & -1 \end{array} \right)$$

Affine shifts.

$$\nu_x = (-3, -2, -2, 0), \quad \nu_p = (2, 0, 0, 0).$$

Interpretation: ν_p shifts the overall $(-q^{1/2})$ exponent; ν_x shifts the η -monomial (trivially, here, since there are no η). Both are intrinsic to the triangulation.

5 Step 4 — Applying the formula to one sector

Before listing the full sector grid, walk through the single sector $(m_{\text{ext}}, e_{\text{ext}}) = (1, 0)$ — the display label $\mathcal{I}_{m022}(0\alpha - \frac{1}{2}\beta)$ — in full.

(a) Build κ . $\kappa = (m_{\text{ext}}, \mathbf{0}^{n-r}, e_{\text{ext}}, \mathbf{e}_{\text{int}}) = (1, 0, 0, 0, 0, e_0, e_1, e_2)$, with $\mathbf{e}_{\text{int}} = (e_0, e_1, e_2) \in (\frac{1}{2}\mathbb{Z})^3$ to be summed over.

(b) Local charges via $g_{\text{NZ}}^{-1}\kappa$.

$$g_{\text{NZ}}^{-1} = \left(\begin{array}{cccc|cccc} 0 & -1 & -1 & -1 & -1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ -1 & -1 & -1 & -1 & -1 & -1 & 0 & -1 \\ \hline 1 & 2 & 3 & 2 & 3 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & 1 & 1 & 2 & 0 & 0 \\ 1 & 2 & 1 & 1 & 1 & 0 & 2 & 0 \end{array} \right)$$

Reading off rows 1:n and $n+1:2n$ and specialising to $m_{\text{ext}} = 1, e_{\text{ext}} = 0$:

$$\begin{array}{ll} \Delta_1 : & m_1 = -e_2, & e_1 = 1 + 2e_0, \\ \Delta_2 : & m_2 = 1 + 2e_2, & e_2 = e_0 + e_1 - 2e_2, \\ \Delta_3 : & m_3 = -1 - e_1, & e_3 = 1 + 2e_0, \\ \Delta_4 : & m_4 = -1 - e_0 - e_2, & e_4 = 1 + 2e_1. \end{array}$$

(c) Integrality constraints. All m_a, e_a must be integers. Since $2e_k$ is always integer for $e_k \in \frac{1}{2}\mathbb{Z}$, the e_a rows are automatically integer. The m_a rows force $e_2 \in \mathbb{Z}$, $e_1 \in \mathbb{Z}$, $e_0 + e_2 \in \mathbb{Z}$, hence $\mathbf{e}_{\text{int}} \in \mathbb{Z}^3$ (not a strict half-step). The internal sum therefore runs over ordinary integer lattice points rather than the full $(\frac{1}{2}\mathbb{Z})^3$.

(d) Prefactor and result. Each admissible $\mathbf{e}_{\text{int}}$ contributes

$$(-q^{1/2})^{\mathbf{m} \cdot \nu_p - \mathbf{e} \cdot \nu_x} \prod_{a=1}^4 \mathcal{I}_{\Delta}(m_a, e_a).$$

Summing over $\mathbf{e}_{\text{int}} \in \mathbb{Z}^3$ and truncating at q^{10} :

$$\mathcal{I}_{m022}(0\alpha - \frac{1}{2}\beta) = -q + 3q^3 + 5q^4 + q^5 - 16q^6 - 37q^7 - 57q^8 - 53q^9 + 9q^{10} + O(q^{11}).$$

6 Step 5 — All computed sectors

The app displays a canonical grid of 5^r sectors per cusp. For $m022$ ($r = 1$) the grid has 5 entries, of which 4 are non-zero. Truncation: $q^{20/2}$, $N_{\text{max}} = 10$. In each row the argument on the left is $e\alpha - \frac{m}{2}\beta$ as in the in-app index table.

$$\begin{aligned}
\mathcal{I}_{m022}(0\alpha + 0\beta) &= 1 - q - q^2 + 4q^3 + 6q^4 + 4q^5 - 16q^6 - 41q^7 - 68q^8 - 73q^9 - 9q^{10} + O(q^{11}), \\
\mathcal{I}_{m022}(0\alpha - \tfrac{1}{2}\beta) &= -q + 3q^3 + 5q^4 + q^5 - 16q^6 - 37q^7 - 57q^8 - 53q^9 + 9q^{10} + O(q^{11}), \\
\mathcal{I}_{m022}(\tfrac{1}{2}\alpha + 0\beta) &= 0, \\
\mathcal{I}_{m022}(0\alpha - \beta) &= 2q^3 + q^4 - 2q^5 - 14q^6 - 24q^7 - 28q^8 - 14q^9 + 47q^{10} + O(q^{11}), \\
\mathcal{I}_{m022}(\alpha + 0\beta) &= -q + 3q^3 + 5q^4 + q^5 - 14q^6 - 35q^7 - 54q^8 - 49q^9 + 9q^{10} + O(q^{11}).
\end{aligned}$$

Observation: Because the sector argument is linear in the signed charges $(e, -m/2)$, the Weyl condition $\mathcal{I}_{m022}(e\alpha - \frac{m}{2}\beta) = \mathcal{I}_{m022}(-e\alpha + \frac{m}{2}\beta)$ is the statement that flipping the sign of every component gives the same q -series. With no hard edges, there is no accompanying $\eta^{\pm n W_a}$ shift to cancel.

7 Step 6 — Weyl-symmetry check

For a general manifold the Weyl-shifted index

$$f(\eta_j; m, e) = \eta_j^{\sum_I a_{j,I} e_I + b_{j,I} m_I} \mathcal{I}_X(e\alpha - \tfrac{m}{2}\beta)$$

is required to satisfy $f(m, e) = f(-m, -e)$ for each hard edge j , together with a pair of integrality conditions on (a_j, b_j) and an adjoint-character projection fixed to -1 per filled cusp.

For **m022** the hard-edge set is empty, so the Weyl vectors **a**, **b** are 0-dimensional:

j	a_j	b_j	$a_j \in \mathbb{Z}?$	$2b_j \in \mathbb{Z}?$
—	—	—	✓ (vacuous)	✓ (vacuous)

Sector-pair symmetry. 4/4 non-zero sectors in the 5-entry display grid satisfy $\mathcal{I}_{m022}(e\alpha - \frac{m}{2}\beta) = \mathcal{I}_{m022}(-e\alpha + \frac{m}{2}\beta)$ ✓.

Adjoint projection. With no η integration, the adjoint-character step reduces to $\sum_e c_e \chi_{\text{adj}}(e) = -1$, where c_e is the q^1 coefficient of $\mathcal{I}_{m022}(e\alpha + 0\beta)$. For **m022**:

e ($= \alpha$ -coeff)	c_e
$-\frac{1}{2}$	0
0	-1
$+\frac{1}{2}$	0

Projected value: -1 (expected -1) ✓.

8 Step 7 — Dehn filling, worked example

Fill the single cusp along the user slope $P_{\text{user}}\alpha + Q_{\text{user}}\beta$ with $(P_{\text{user}}, Q_{\text{user}}) = (5, 2)$.

(a) Non-closable cycle search

Search range $P \in [-10, 10]$, $Q \in [1, 10]$ [notional range]. One NC cycle returned:

$$\gamma = P_{\text{nc}}\alpha + Q_{\text{nc}}\beta, \quad (P_{\text{nc}}, Q_{\text{nc}}) = (1, 1). \quad [\text{notional}]$$

(b) Complementary basis vector δ

Solve $R Q_{\text{nc}} - P_{\text{nc}} S = 1$ by extended Euclid: $(R, S) = (0, -1)$, giving $\delta = 0 \cdot \alpha + (-1) \cdot \beta$. Verify $\det \begin{pmatrix} P_{\text{nc}} & Q_{\text{nc}} \\ R & S \end{pmatrix} = (1)(-1) - (1)(0) = -1 \neq 0$ ✓.

(c) User slope in the (γ, δ) basis

Expand $P_{\text{user}}\alpha + Q_{\text{user}}\beta = p\gamma + q\delta$:

$$\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} S & -R \\ -Q_{\text{nc}} & P_{\text{nc}} \end{pmatrix} \begin{pmatrix} P_{\text{user}} \\ Q_{\text{user}} \end{pmatrix} = \begin{pmatrix} -5 \\ -3 \end{pmatrix}.$$

The filled index is displayed with the user slope (not the transformed one) in the subscript, as $\mathcal{I}_{(\text{m022})_{5\alpha+2\beta}}$; the transformed slope $p/q = (-5)/(-3)$ enters the HJ step next.

(d) Hirzebruch–Jung continued fraction

$p/q = (-5)/(-3)$. The HJ algorithm returns $[2, 2, 2]$, length $\ell = 3$. [notional; `hj_continued_fraction` would be run on real (p, q)]

(e) Basis change on g_{NZ}

`apply_cusp_basis_change` replaces the $(\alpha, \beta/2)$ rows of g_{NZ} with

$$\text{new-position} = P_{\text{nc}} \alpha + 2Q_{\text{nc}} (\beta/2), \quad \text{new-momentum} = a \alpha + b (\beta/2),$$

where (a, b) satisfy the Bézout identity $P_{\text{nc}} b - 2Q_{\text{nc}} a = 1$. The result:

$$g'_{\text{NZ}} = ([8 \times 8 \text{ integer matrix, printed in full in the real report}]), \quad \nu'_x = (\text{updated}), \quad \nu'_p = (\text{updated}).$$

(f) Physical Weyl vectors and edge compatibility

With 0 hard edges, $\mathbf{a}_{\text{phys}}$ and $\mathbf{b}_{\text{phys}}$ are empty 0-vectors; no edge is collapsed; all hard edges (vacuously) integer-compatible. ✓

(g) Kernel product & contraction

HJ kernel assembly for $[2, 2, 2]$ produces a truncated polynomial with ~ 17 non-zero terms at the chosen q -order. Contracting against the pre-computed $\mathcal{I}_{\text{m022}}$ and collecting residual cusp- η fugacities η^{nV_0} :

$$\mathcal{I}_{(\text{m022})_{5\alpha+2\beta}} = 1 + (-2 - \eta^{2V_0} - \eta^{-2V_0})q + (\eta^{4V_0} + \eta^{-4V_0} - 4)q^2 + \dots + O(q^{11}). \quad [\text{notional coefficients}]$$

(For a manifold with hard edges, terms of the form $\eta^{\pm nW_a} \eta^{nV_0}$ would also appear; cf. the $(5_2)_{\alpha+2\beta}$ series in the in-app display.)

Appendix — Tetrahedron index definition

$$I_t(m, e) = \sum_{k=\max(0, -e)}^{\infty} \frac{(-1)^k q^{k(k+1)/2 - (2k+e)m/2}}{\prod_{j=1}^k (1 - q^j) \prod_{j=1}^{k+e} (1 - q^j)}, \quad \mathcal{I}_{\Delta}(m, e) = \begin{cases} (-q^{1/2})^m I_t(-m - e, m), & m + e \geq 0, \\ I_t(m, e), & m + e < 0. \end{cases}$$